

Introduction to Computational Thinking and Python Programming

Instructor: Sarom Leang, PhD (Teacher/Professor)

Specialist: Ravi Teja Talluri

Innovation Hall 203



Pronouns

- Instructor: Sarom Leang, Ph.D. (Teacher/Professor)
- Specialist: Ravi Teja Talluri



Schedule

- 10:00 AM – 10:15 AM (Homeroom)
- 10:15 AM – 11:35 AM (G1)
- 11:40 AM – 12:35 PM (Lunch)
- 12:40 PM – 02:00 PM (G2)



Student Expectations

- **NO FOOD**
- **NO DRINKS** (on the table)
- Be respectful to individuals and property
- Be open to learning
- Be open to not understanding
- Be patient with yourself
- Ask questions
- Explore
- **Embrace failure**



Course Overview (cont.)

Exploratory Topics

- ~~Session #1 – November 2, 2024~~
 - ~~Entrance Survey~~
 - ~~How Computers Work~~
- ~~Session #2 – December 7, 2024~~
 - ~~Parallel Computing~~
- Session #3 – February 1, 2025
 - Generative Artificial Intelligence
- Session #4 – March 22, 2025
 - Quantum Computing
- Session #5 – April 12, 2025
 - Future of Computing
 - Exit survey

Topics

- Algorithms and Problem Solving
- Debugging and Troubleshooting
- Loops
- Algorithms and Syntax
- Variables and Conditionals
- Variable Arithmetic
- Conditionals (If/Else)
- Compound Conditionals





Happy Lunar New Year



Department of Energy Secretary Honors Award
Exascale Computing Project

Recap: Session 1

- How do computers work?
- Data vs. Information
- What is computer programming?
- Ozaria



Recap: Session 2

- High performance computing and parallel programming
- FLOPs – Floating point operations per second
- 1 ExaFLOP
- Supercomputing applications
- Computational thinking – 3D puzzle
 - **Problem** - connect the most number of puzzle pieces together
 - **Decomposition** - separating, sorting, and connecting pieces
 - **Pattern recognition** – connect pieces in numeric order for bottom, middle and top
 - **Abstraction** - pick up a piece, identify piece number, connect neighboring pieces
 - **Algorithm design** – separate, sort, connect, combine, count
- Challenges in high performance computing



NEWS



Technology in the News

DeepSeek-R1 LLM (Large Language Model)

- Open-source AI model developed by a Chinese company
- Performs better than OpenAI's o1 model
- Training cost for DeepSeek-R1 estimated around \$5-6 million
- Training cost for OpenAI's models are well over hundreds of million
- Reported DeepSeek-R1 was trained using mid-range NVIDIA GPUs
- Export ban on NVIDIA top-range GPUs (Hopper H100s)
- Triggered technology stock sell off



Generative AI



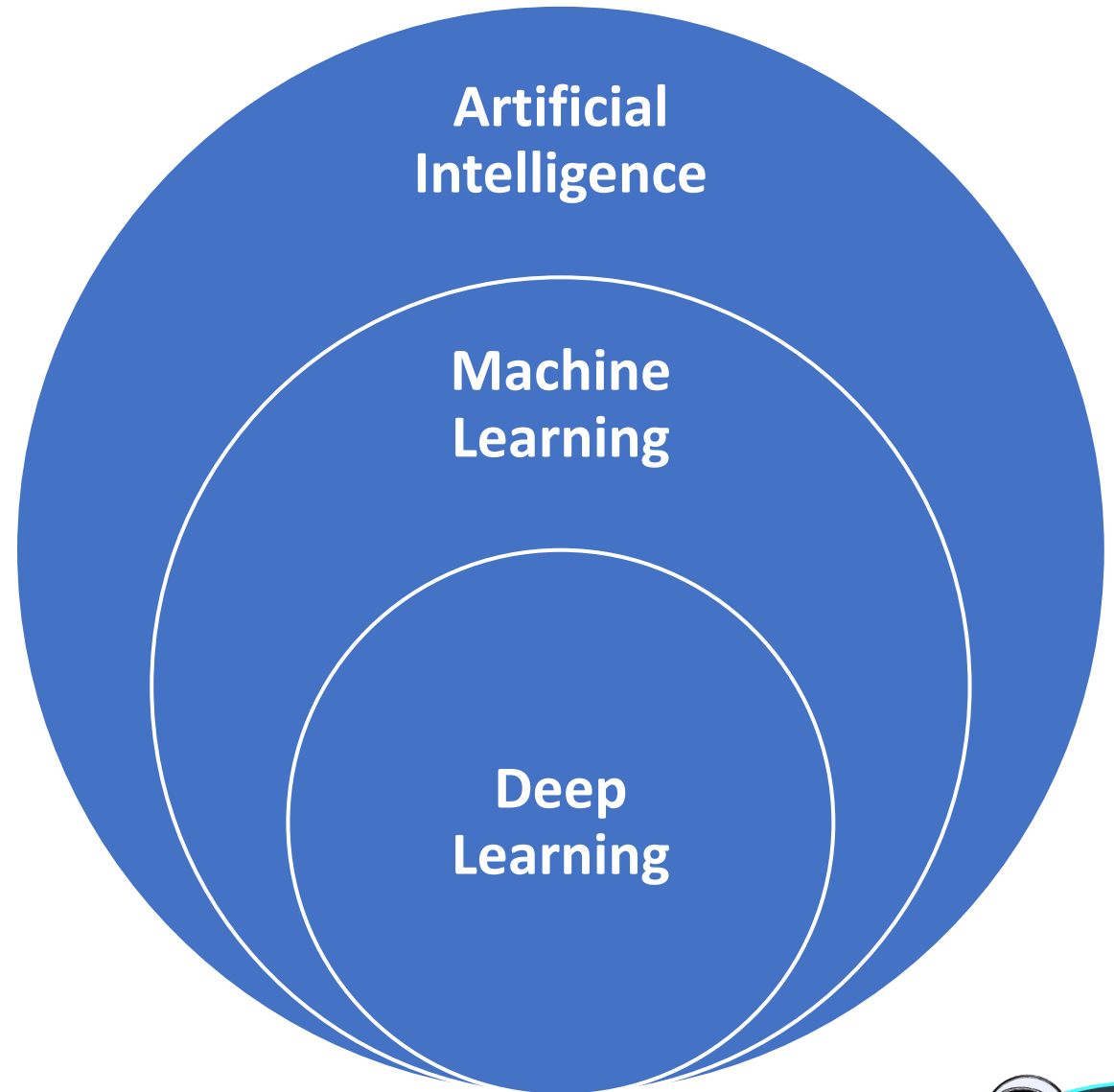
Generative AI

- **Artificial Intelligence** – technology designed to **perform tasks** that would typically **require human intelligence**
- **Generative** – creating **new data based on existing knowledge**
- **Generative AI** – is a technology designed to create new data based on learned patterns (**statistical pattern recognition**)
- **Data**
 - Text
 - Images
 - Music
 - Videos



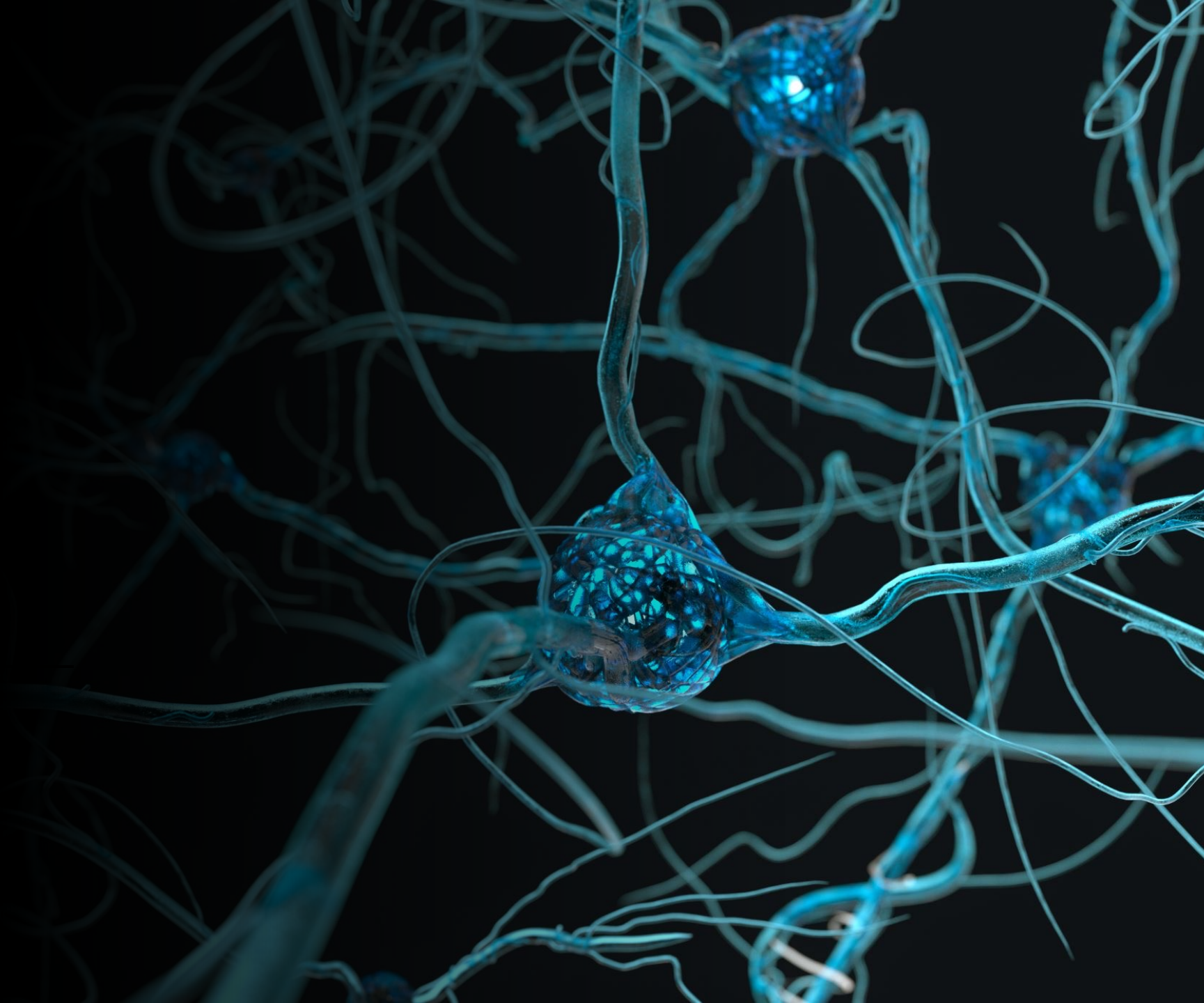
AI vs. ML vs. DL

- AI (artificial intelligence) - Creation of **intelligent machines** that can **perform tasks** that would typically **require human intelligence** to complete
- ML (machine learning) - teaching computers to **learn** and **make decisions** on their own
- DL (deep learning) - type of ML that uses multi-layer neural networks to **analyze** and **interpret data** (**recognize patterns**)



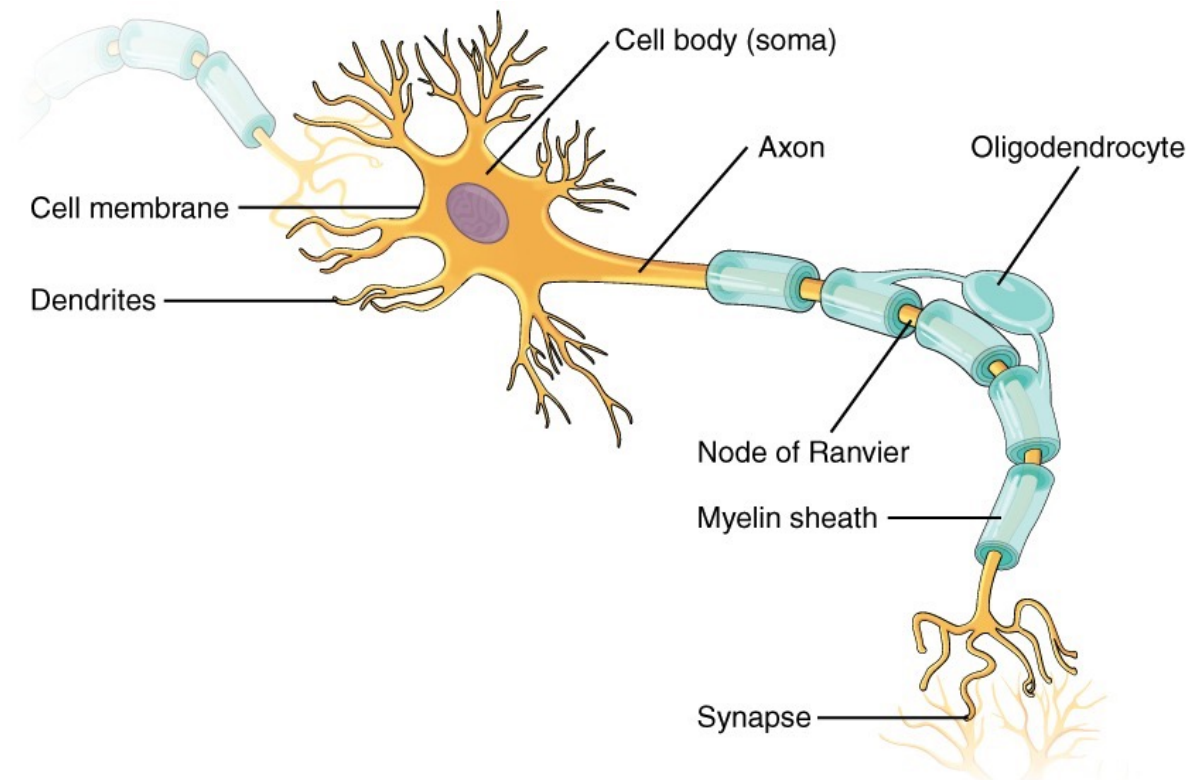


Neural Networks



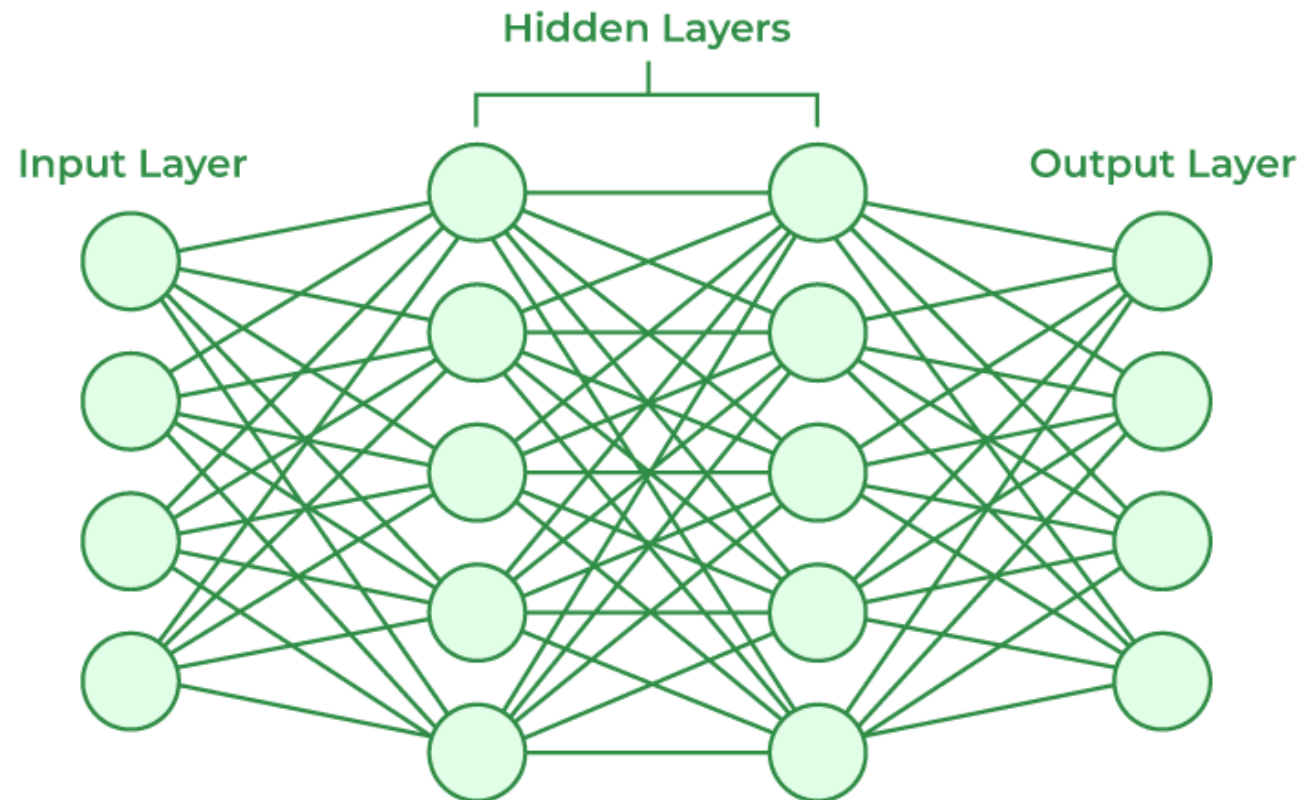
Neural Networks (Neuroscience)

- Networks of nerve cells (**neurons**) in the brain and nervous system that are interconnected and **communicate through synapses** to **process information, control bodily functions, and enable behaviors**



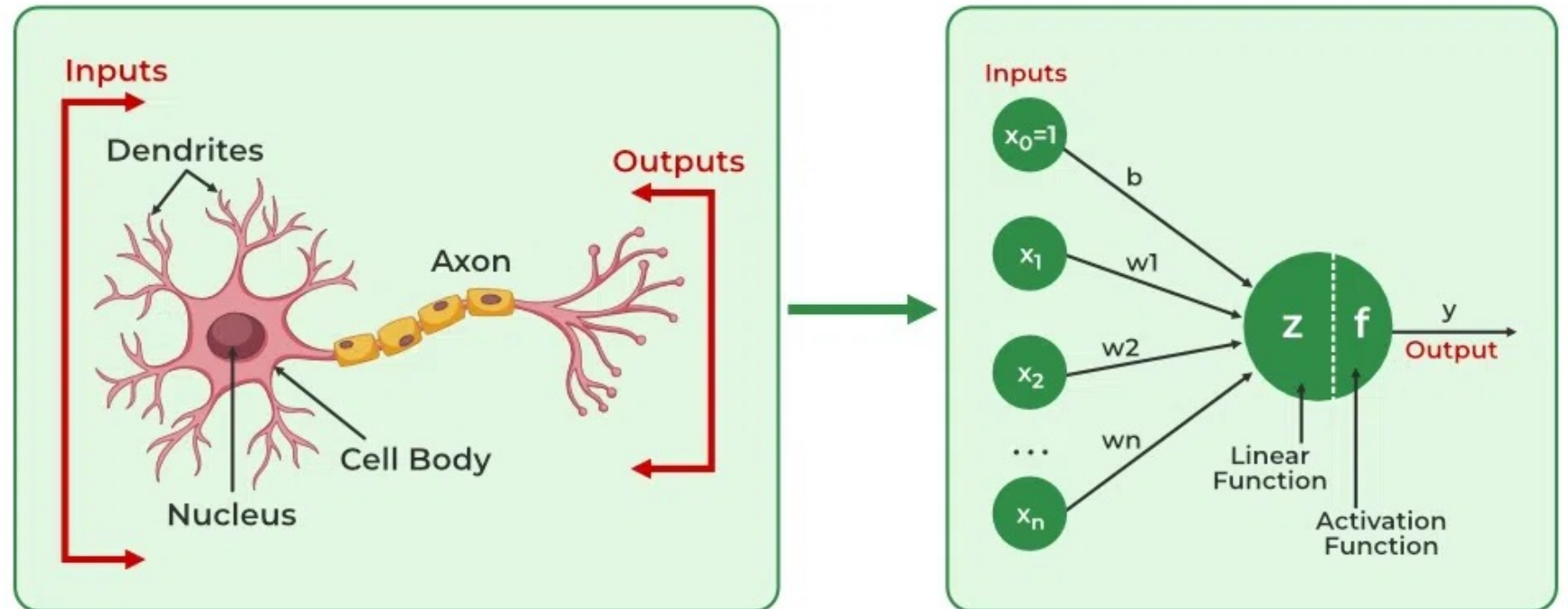
Neural Networks (Computer Science)

- Also referred to as artificial neural networks, loosely modeled on the organization of the human brain, form the basis for many artificial intelligence systems

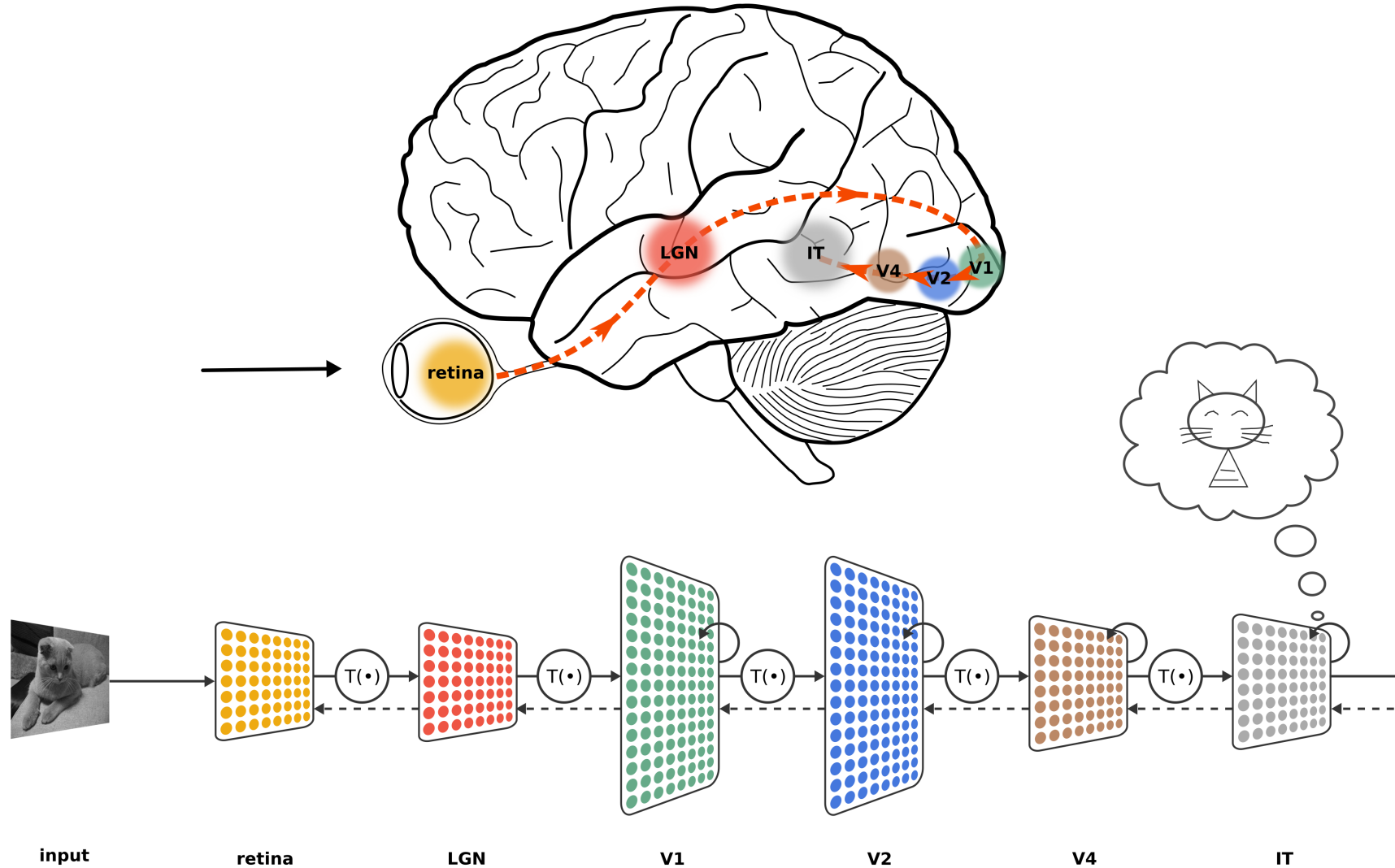


Neural Networks

Biological	Artificial
Dendrite	Inputs
Nucleus	Nodes
Synapses	Weights
Axon	Output



Neural Networks

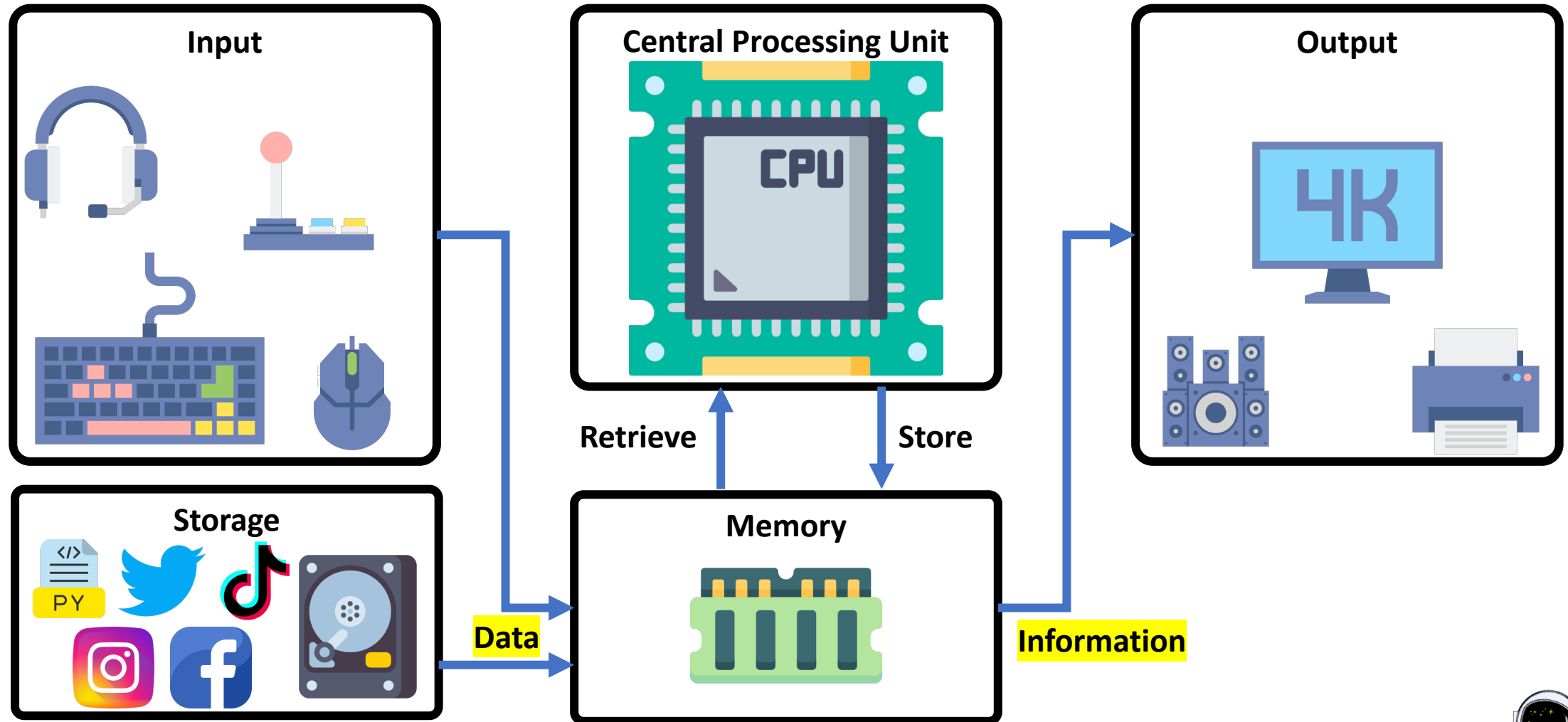


Generative AI

- A type of artificial intelligence that can **create new content that is similar to the original data it was trained on**
- **Learns patterns and relationships in data, and uses that knowledge to generate new content**



Data -> Information



Data -> Information -> Knowledge

Data (raw data)	Information (processed and organized)	Knowledge (insight gained)
Census records	Demographics	What will be the average age of the population in 2040?
Weather measurements	Average temperature, rainfall, etc. in a region over a given time range	How will the weather be next week, month, year?
Financial records	Growth rate of a company	Where will this company be in the next five years?
Patient records	Incident rate of a disease	When will COVID-19 end?

Generative AI Tools

- Text – [ChatGPT](#), Claude, Gemini
- Image – [DALL-E](#), Midjourney, Stable Diffusion, Adobe Firefly, Imagen
- Music – AIWA, Soundful, Mubert, Beatoven, Suno AI, [MusicGPT](#)
- Videos – [Sora](#), Runway, Descript, Capsule



OpenAI ChatGPT, Anthropic Claude, and Google Gemini

- Interact with these AI model by providing prompts
- Prompts may describe a task being asked of the model
- Prompts are inputs to the model
- Generative content are the outputs from the model
- The performance (quality of the output) of these models are dependent on the inputs
- **Prompt Engineering** - creating specific prompts or inputs that can guide the generative AI model to produce output that is relevant and coherent

Example Prompts for Text Generation

- What is the difference between data, information, and knowledge?

- Explain this python code:

```
r=int(input("Enter upper limit: "))
```

```
for a in range(2,r+1):
```

```
    k=0
```

```
    for i in range(2,a//2+1):
```

```
        if(a%i==0):
```

```
            k=k+1
```

```
if(k<=0):
```

```
    print(a)
```



Example Prompts for Text Generation

- What is an ideal time within business hours to host a virtual conference with attendees in:
 - Ames, Iowa
 - Arlington, Virginia
 - Canberra, Australia?
- Adjust your suggestions with business hours extending to 6PM local time.
- These meetings will take place in September.



Example Prompts for Text Generation

- Write a positive Yelp review of Café Renaissance in Vienna, Virginia. Highlighting their grilled halibut dish in almond and lemon butter sauce
 - More witty and colorful
 - End with: It was a wonderful way to spend Valentines Day
 - Also highlight the Escargot à la Bourguignonne appetizer which was cooked in garlic butter parsley and brandy
- Generate an image for the review



Example Prompts for Image Generation

- Create an image of a koala bear crossing the Delaware River
 - While wearing a pink hoody
 - With a Sony Walkman
- Generate an image of the Oregon Trail passage but replace the wagons with low riders
- Generate an image of a female US President with Cambodian roots
- A painting of Michael Jordan slam dunking in the air while eating an avocado
- Synthwave George Mason University logo
- A painting of dark-skinned female Santa having dinner with dark-skinned female Jesus



Example Prompts for Movie Generation

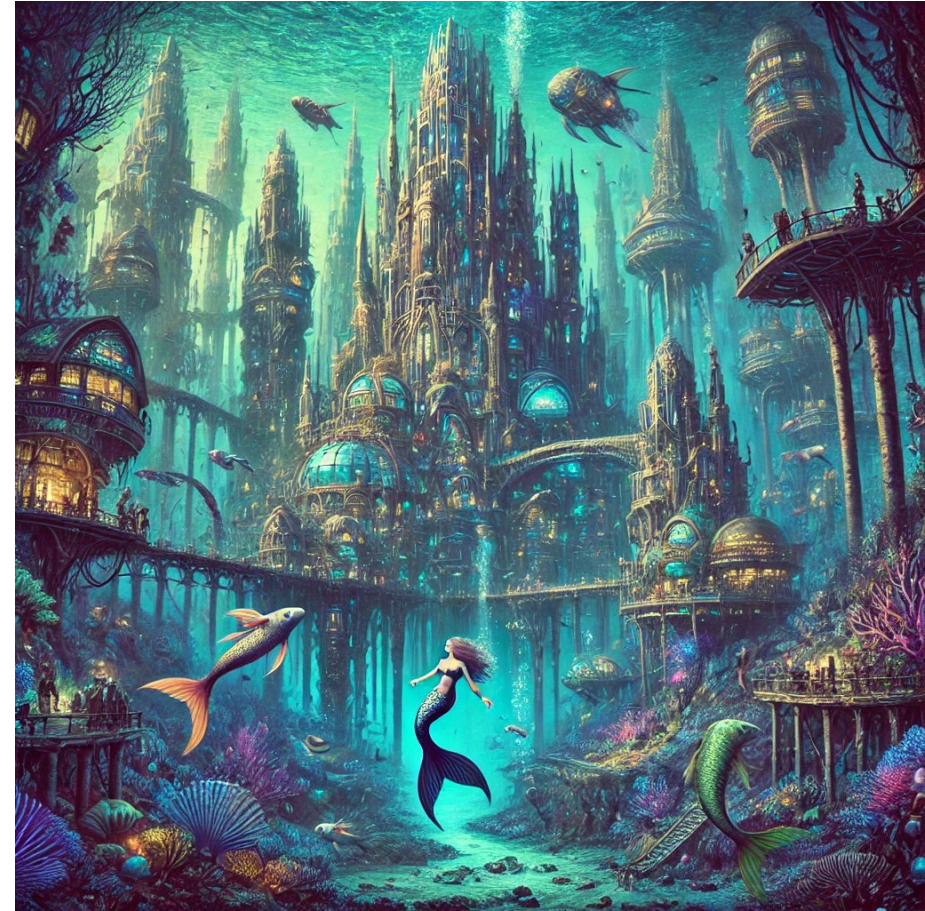
- Oxen wagon race at the Daytona speedway
- Koalas playing 3 on 3 basketball
- Lunar new year dragon dance in front of the Washington monument
- Jesus on a Segway in New York City during rush hour doing food delivery



Generative AI Inception

Ask ChatGPT to generate a prompt for DALL-E

Create an image of a fantastical underwater city, with towering structures and bustling streets, where mermaids and other sea creatures live and work alongside human-like beings. The city should have a distinct aesthetic, with colors and architectural elements that evoke the feeling of being submerged in the depths of the ocean. The image should also include unique transportation methods, such as aquatic vehicles or creatures that are used for travel.



Generative AI Timeline

	Pre-2020	2020	2022	2023	2025	2030
Text	Spam detection Translation Basic Q&A	Basic copywriting First drafts	Longer form Second drafts	Vertical fine-tuning gets good (scientific papers)	Final drafts better than the human average	Final drafts better than professional writers
Code	1-line auto-complete	Multi-line generation	Longer form Better accuracy	More languages More verticals	Text to product (draft)	Text to product (final) better than full-time developers
Images			Art Logo Photography	Mock-ups (product design, architecture, etc.)	Final drafts (product design, architecture, etc.)	Final drafts better than professional artists, designers, photographers
Video/3D/ Gaming			First attempts at 3D/video models	Basic/first draft video and 3D files	Second drafts	AI Roblox Video games and movies are personalized dreams

<https://ig.ft.com/generative-ai/>



Generative AI Limitations

- Generative AI can produce content that is **incorrect** or **incoherent (hallucinate)**
- Generative models don't understand the content they generate. They **generate responses based on statistical patterns rather than comprehension**, which means they can't answer questions that require understanding or reasoning
- Generative AI can perpetuate **biases** present in the training data
- Generative models can be **verbose** and tend to **over generate** content
- Same prompt may produce **inconsistent** responses using the same model



Generative AI and Academic Code of Conduct

- Cheating or violating academic code of conduct using generative AI is unethical and has serious consequences
- Disciplinary action may include failing the assignment or course, or even expulsion from school
- Long-term consequences may include damaging a student's reputation and future academic and career prospects
- Use generative AI responsibly and in accordance with the academic code of conduct

